Appendix S1

Spatiotemporal Graph Neural Networks for Neighborhood-Level Influenza Forecasting in Boston

Contents. Appendix S1 (models, tuning, data notes, and design experience on the development season); Tables S1–S13; Figures S1–S5.

Models

Table S1 lists every model in the evaluation. All learned models were tuned with the same budget in every test season and averaged over 10 training runs, except in Chicago, where every model ran at default settings and the LSTM and gradient-boosted trees were trained once; all models except the FluSight baseline received forecast intervals from the same shared procedure described in the main text.

Table S1: Models Compared

Model	What it does	Inputs
Graph neural network	Causal temporal convolutions over each neighborhood's recent weeks, graph convolution over the neighborhood map plus a learned set of extra links, and a learned correction to a blend of the current week and a seasonal curve	16 weeks of rates, 6 weather, 8 demographic measures, map
Same model, map removed	As above with no links between neighborhoods and no surrounding areas	Same, no map
LSTM	Long short-term memory network reading all neighborhoods' series jointly	All series, no map
Cola-GNN	Published influenza graph model that learns connections from the data;[1] reimplemented	All series
Gradient-boosted trees	Regression trees pooled across neighborhoods	Same columns as graph model, no map
Seasonal ARIMA	Per-neighborhood ARIMA with the annual cycle as harmonic regressors	Own history
Seasonal blend, no learning	The graph model's baseline (current week blended with the seasonal curve) with the network switched off	Own history
FluSight baseline	Latest value carried forward, ranges from past changes over the forecast horizon	Own history
Persistence	Next week equals this week	Own history
Climatology	Seasonal curve for the target week	Own history
Same week last year	The value in the same week one year earlier	Own history
Graph attention network[a]	Spatial graph attention over 52 weeks of history, no temporal layer[2,3]	52 weeks, map
Dual-topology GCN[a]	Separate geographic and correlation graphs with temporal convolutions[3]	52 weeks, 2 maps

Abbreviations: ARIMA, autoregressive integrated moving average; GCN, graph convolutional network; LSTM, long short-term memory network.

[a] The comparison and reference models of Luo et al.,[3] run at 1 week ahead only, as in that paper.

Comparison models: ARIMA and LSTM

[Coauthor: describe the ARIMA comparison model here (about 120–170 words).]

[Coauthor: describe the LSTM comparison model here (about 130–180 words).]

Equal-budget tuning

For each learned model, test season, and horizon, 12 configurations were trained once: the model's default and 11 drawn at random from the space in Table S2. The configuration with the lowest mean absolute error on the validation weeks immediately before the test season was retrained with 10 random seeds and scored on the test season. The chosen configurations are recorded in `Code/sweep/tuned_configs.json`.

Table S2: Tuning Search Spaces

Model	Values searched
Graph neural network (with and without map)	learning rate {0.001, 0.003, 0.01}; weight decay {0.0003, 0.001, 0.003, 0.01}; dropout {0.1, 0.25, 0.4}; channels {16, 24, 32}; epochs {80, 120, 200}
LSTM	learning rate {0.0003, 0.001, 0.003}; weight decay {0.00001, 0.0001, 0.001}; dropout {0, 0.1, 0.25}; hidden size {32, 64, 128}; lookback {8, 16, 26} weeks; epochs {80, 120, 200}; target {level, blend}
Gradient-boosted trees	learning rate {0.01, 0.03, 0.1}; maximum depth {2, 3, 4, 6}; minimum child weight {1, 4, 8}; row subsample {0.6, 0.8, 1.0}; column subsample {0.6, 0.8, 1.0}; L2 penalty {0.5, 2, 8}; target {climatology, level, blend}
Cola-GNN	learning rate {0.0003, 0.001, 0.003}; weight decay {0, 0.0005, 0.005}; dropout {0.1, 0.2, 0.4}; hidden size {16, 32, 64}; lookback {12, 20, 26} weeks; epochs {100, 150, 200}
Graph attention network	learning rate {0.001, 0.003, 0.01}; weight decay {0.00001, 0.0001, 0.001}; dropout {0.1, 0.2, 0.4}; hidden size {32, 64}; heads {4, 8}; epochs {100, 200}
Dual-topology GCN	learning rate {0.001, 0.003, 0.01}; weight decay {0.00001, 0.0001, 0.001}; epochs {100, 200}; fusion {concatenate, mean, maximum}

Abbreviations: GCN, graph convolutional network; LSTM, long short-term memory network.

Full results by season

Table S3: Mean Weighted Interval Score by Model, Season, and Horizon, 14 Neighborhoods, Boston, Massachusetts, 2022–2025

	2022–2023			2023–2024			2024–2025		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	5.56	7.70	10.95	4.70	5.07	5.53	5.14	6.64	8.18
Same model, map removed	6.23	7.83	10.86	5.04	5.37	5.98	5.79	6.95	9.20
LSTM	5.50	7.67	11.29	4.85	5.85	6.84	5.20	6.96	9.33
Cola−GNN	6.01	10.37	10.97	4.79	5.29	6.43	5.24	7.17	8.92
Gradient−boosted trees	6.29	8.09	11.31	5.28	5.61	6.34	6.68	8.27	9.49
Seasonal ARIMA	6.29	8.02	10.94	5.43	5.75	5.91	7.84	9.22	10.60
Seasonal blend, no learning	6.26	7.95	11.30	5.63	6.08	6.44	5.97	7.84	10.96
FluSight baseline	6.65	8.55	12.50	5.83	6.52	7.68	6.69	9.41	14.17
Persistence	6.51	8.30	12.38	5.86	6.57	7.87	6.36	8.95	14.07
Graph attention network[a]	6.32			5.08			5.02		
Dual−topology GCN[a]	11.36			5.38			6.45		
Climatology	13.25	13.25	13.21	6.59	6.60	6.60	11.10	11.10	11.10
Same week last year	12.19	12.19	12.19	13.71	13.79	13.79	10.61	10.61	10.61

Values are emergency department visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Run at 1 week ahead only.

Table S4: Mean Weighted Interval Score by Model and Horizon, 17 Areas, Columbus, Ohio, June 2024–May 2025

Model	1 wk	2 wk	4 wk
Graph neural network	4.06	5.03	6.31
Same model, map removed	4.70	5.37	6.39
LSTM	5.13	7.13	6.57
Cola−GNN	6.03	6.75	7.12
Gradient−boosted trees	6.96	5.47	6.59
Seasonal ARIMA	5.55	5.84	6.50
Seasonal blend, no learning	5.05	5.68	6.44
FluSight baseline	4.84	6.10	8.20
Persistence	4.67	5.89	8.45
Graph attention network[a]	7.11		
Dual−topology GCN[a]	8.71		
Climatology	6.75	6.75	6.75
Same week last year	5.75	5.75	5.75

Values are emergency department visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Run at 1 week ahead only.

Table S5: Mean Weighted Interval Score by Model, Season, and Horizon, 58 ZIP Codes, Chicago, Illinois, 2022–2026

	2022–2023			2023–2024			2024–2025			2025–2026		
Model	1	2	4	1	2	4	1	2	4	1	2	4
Graph neural network	1.35	1.42	1.52	1.09	1.13	1.26	1.09	1.14	1.27	1.00	1.07	1.18
Same model, map removed	1.26	1.38	1.52	1.14	1.20	1.28	1.17	1.24	1.34	1.08	1.15	1.22
LSTM	1.60	1.62	1.60	1.12	1.18	1.30	1.14	1.43	1.60	1.07	1.16	1.31
Gradient−boosted trees	1.40	1.50	1.66	1.16	1.20	1.28	1.22	1.26	1.40	1.12	1.18	1.23
Seasonal ARIMA	1.31	1.37	1.52	1.17	1.19	1.21	1.25	1.27	1.35	1.10	1.15	1.20
Seasonal blend, no learning	1.37	1.44	1.55	1.18	1.20	1.24	1.28	1.32	1.36	1.15	1.20	1.27
Persistence	1.44	1.52	1.68	1.33	1.39	1.48	1.35	1.42	1.52	1.25	1.32	1.50
Climatology	1.69	1.69	1.69	1.28	1.28	1.28	1.51	1.51	1.51	1.26	1.27	1.27
Same week last year	1.76	1.76	1.76	1.63	1.63	1.63	1.57	1.57	1.57	1.47	1.47	1.47

Values are in Chicago Department of Public Health influenza-like illness activity levels (1–10), not rates, so they are not comparable in size with Boston or Columbus; lower is better. Column numbers are weeks ahead. No Chicago season was used in design, so all 4 are unseen. Every model ran at its default settings (no equal-budget tuning), and the LSTM and gradient-boosted trees were trained once; Cola-GNN and the FluSight baseline were not run.

Week-level comparisons

Each row compares the graph neural network with one model on the weeks of one season. The difference is graph neural network minus the other model, so a negative value favors the graph neural network. Intervals are 95% confidence intervals from a moving-block bootstrap over weeks (blocks of 4 weeks). *P* values are from the 2-sided Diebold–Mariano test with the Harvey–Leybourne–Newbold correction, Holm-adjusted across every model compared in that season and horizon (including the 1-week-only models and an untuned 10-seed LSTM, which are not shown). In Chicago every model ran at default settings (the LSTM and gradient-boosted trees were trained once), and the adjustment covers all Chicago models.

Table S6: Week-Level Differences in Weighted Interval Score, Boston, Massachusetts, 2022–2025

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted *P*
2022–2023	1	Same model, map removed	−0.68 (−1.03, −0.36)	<.001
		LSTM	0.06 (−0.22, 0.34)	.62
		Cola-GNN	−0.45 (−1.32, 0.09)	.30
		Gradient-boosted trees	−0.73 (−1.15, −0.37)	<.001
		Seasonal ARIMA	−0.74 (−1.31, −0.37)	.001
		Seasonal blend, no learning	−0.70 (−1.11, −0.39)	<.001
		FluSight baseline	−1.09 (−2.14, −0.47)	.001
		Persistence	−0.95 (−1.94, −0.31)	.005
2022–2023	2	Same model, map removed	−0.13 (−0.87, 0.60)	>.99
		LSTM	0.03 (−0.78, 0.63)	>.99
		Cola-GNN	−2.67 (−7.23, 0.22)	.95
		Gradient-boosted trees	−0.39 (−0.60, −0.20)	.001
		Seasonal ARIMA	−0.32 (−1.24, 0.57)	>.99
		Seasonal blend, no learning	−0.25 (−1.30, 0.71)	>.99
		FluSight baseline	−0.85 (−3.37, 0.85)	>.99
		Persistence	−0.60 (−3.12, 1.40)	>.99
2022–2023	4	Same model, map removed	0.09 (−0.18, 0.43)	>.99
		LSTM	−0.34 (−3.55, 2.68)	>.99
		Cola-GNN	−0.02 (−2.83, 2.77)	>.99
		Gradient-boosted trees	−0.36 (−0.81, 0.00)	>.99
		Seasonal ARIMA	0.01 (−1.51, 1.81)	>.99
		Seasonal blend, no learning	−0.35 (−2.16, 1.20)	>.99
		FluSight baseline	−1.55 (−6.37, 1.79)	>.99
		Persistence	−1.43 (−6.56, 2.72)	>.99
2023–2024	1	Same model, map removed	−0.34 (−0.48, −0.17)	.009
		LSTM	−0.14 (−0.42, 0.05)	.43
		Cola-GNN	−0.09 (−0.26, 0.05)	.43
		Gradient-boosted trees	−0.57 (−0.80, −0.37)	<.001
		Seasonal ARIMA	−0.73 (−0.95, −0.50)	<.001
		Seasonal blend, no learning	−0.93 (−1.24, −0.66)	<.001
		FluSight baseline	−1.13 (−1.48, −0.84)	<.001
		Persistence	−1.15 (−1.46, −0.88)	<.001
2023–2024	2	Same model, map removed	−0.30 (−0.50, −0.11)	.01
		LSTM	−0.78 (−1.05, −0.51)	<.001
		Cola-GNN	−0.22 (−0.71, 0.18)	.46
		Gradient-boosted trees	−0.54 (−0.77, −0.34)	<.001

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
		Seasonal ARIMA	−0.68 (−0.98, −0.46)	<.001
		Seasonal blend, no learning	−1.01 (−1.49, −0.68)	<.001
		FluSight baseline	−1.45 (−2.07, −0.99)	<.001
		Persistence	−1.50 (−2.03, −1.06)	<.001
2023–2024	4	Same model, map removed	−0.45 (−0.75, −0.17)	.03
		LSTM	−1.31 (−2.28, −0.53)	.03
		Cola-GNN	−0.90 (−2.10, −0.02)	.10
		Gradient-boosted trees	−0.81 (−1.09, −0.54)	<.001
		Seasonal ARIMA	−0.38 (−0.62, −0.10)	.04
		Seasonal blend, no learning	−0.90 (−1.42, −0.49)	.004
		FluSight baseline	−2.15 (−3.22, −1.35)	.001
		Persistence	−2.34 (−3.42, −1.47)	<.001
2024–2025	1	Same model, map removed	−0.65 (−1.25, −0.18)	.009
		LSTM	−0.06 (−0.41, 0.25)	>.99
		Cola-GNN	−0.10 (−0.43, 0.24)	>.99
		Gradient-boosted trees	−1.54 (−3.20, −0.51)	.01
		Seasonal ARIMA	−2.70 (−5.17, −1.10)	.001
		Seasonal blend, no learning	−0.83 (−1.39, −0.41)	<.001
		FluSight baseline	−1.55 (−2.83, −0.67)	.01
		Persistence	−1.22 (−2.41, −0.35)	.04
2024–2025	2	Same model, map removed	−0.31 (−0.57, −0.09)	.08
		LSTM	−0.33 (−0.74, 0.07)	.25
		Cola-GNN	−0.54 (−1.31, 0.06)	.25
		Gradient-boosted trees	−1.63 (−3.53, −0.48)	.19
		Seasonal ARIMA	−2.59 (−4.62, −1.19)	.009
		Seasonal blend, no learning	−1.21 (−1.98, −0.60)	.002
		FluSight baseline	−2.78 (−5.81, −0.86)	.17
		Persistence	−2.31 (−5.12, −0.39)	.19
2024–2025	4	Same model, map removed	−1.01 (−1.82, −0.40)	.06
		LSTM	−1.15 (−2.13, −0.43)	.07
		Cola-GNN	−0.74 (−1.70, −0.02)	.17
		Gradient-boosted trees	−1.31 (−2.69, −0.35)	.17
		Seasonal ARIMA	−2.41 (−3.89, −1.18)	.02
		Seasonal blend, no learning	−2.78 (−4.71, −1.07)	.06
		FluSight baseline	−5.98 (−12.10, −2.04)	.17
		Persistence	−5.88 (−12.08, −1.81)	.17

Table S7: Week-Level Differences in Weighted Interval Score, Columbus, Ohio, 2024–2025

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
2024–2025	1	Same model, map removed	−0.64 (−1.58, −0.06)	.07
		LSTM	−1.07 (−2.81, −0.08)	.07
		Cola-GNN	−1.96 (−4.87, −0.23)	.05
		Gradient-boosted trees	−2.90 (−7.04, −0.26)	.05
		Seasonal ARIMA	−1.49 (−3.34, −0.30)	.02
		Seasonal blend, no learning	−0.98 (−2.52, −0.04)	.07
		FluSight baseline	−0.78 (−1.47, −0.07)	.05
		Persistence	−0.60 (−1.13, 0.16)	.07

2024–2025	2	Same model, map removed	−0.35 (−1.14, 0.22)	.63
		LSTM	−2.10 (−4.72, −0.33)	.21
		Cola-GNN	−1.73 (−3.91, −0.30)	.23
		Gradient-boosted trees	−0.44 (−1.29, 0.12)	.63
		Seasonal ARIMA	−0.81 (−2.10, 0.02)	.63
		Seasonal blend, no learning	−0.66 (−1.93, 0.14)	.63
		FluSight baseline	−1.07 (−2.64, 0.23)	.63
		Persistence	−0.87 (−1.90, 0.27)	.63
2024–2025	4	Same model, map removed	−0.08 (−0.39, 0.21)	.94
		LSTM	−0.26 (−0.52, −0.10)	.25
		Cola-GNN	−0.80 (−1.44, −0.35)	.10
		Gradient-boosted trees	−0.28 (−0.72, 0.09)	.89
		Seasonal ARIMA	−0.19 (−0.44, 0.07)	.88
		Seasonal blend, no learning	−0.13 (−0.41, 0.13)	.94
		FluSight baseline	−1.88 (−5.62, 1.05)	.94
		Persistence	−2.14 (−4.48, −0.18)	.40

Table S8: Week-Level Differences in Weighted Interval Score, Chicago, Illinois, 2022–2026 (Activity Levels, Default Settings)

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
2022–2023	1	Same model, map removed	0.09 (−0.03, 0.20)	.10
		LSTM	−0.26 (−0.81, 0.13)	.32
		Gradient-boosted trees	−0.05 (−0.34, 0.17)	>.99
		Seasonal ARIMA	0.04 (−0.05, 0.13)	.91
		Seasonal blend, no learning	−0.02 (−0.24, 0.17)	>.99
		Persistence	−0.09 (−0.14, −0.03)	<.001
2022–2023	2	Same model, map removed	0.04 (−0.14, 0.17)	>.99
		LSTM	−0.20 (−0.72, 0.16)	>.99
		Gradient-boosted trees	−0.09 (−0.41, 0.15)	>.99
		Seasonal ARIMA	0.05 (−0.03, 0.15)	>.99
		Seasonal blend, no learning	−0.02 (−0.24, 0.17)	>.99
		Persistence	−0.10 (−0.20, 0.00)	.27
2022–2023	4	Same model, map removed	0.00 (−0.11, 0.09)	>.99
		LSTM	−0.08 (−0.41, 0.20)	>.99
		Gradient-boosted trees	−0.14 (−0.52, 0.18)	>.99
		Seasonal ARIMA	−0.00 (−0.09, 0.10)	>.99
		Seasonal blend, no learning	−0.03 (−0.30, 0.21)	>.99
		Persistence	−0.16 (−0.30, 0.01)	.66
2023–2024	1	Same model, map removed	−0.05 (−0.09, −0.02)	<.001
		LSTM	−0.03 (−0.08, 0.00)	.07
		Gradient-boosted trees	−0.07 (−0.13, −0.03)	<.001
		Seasonal ARIMA	−0.08 (−0.12, −0.05)	<.001
		Seasonal blend, no learning	−0.09 (−0.14, −0.05)	<.001
		Persistence	−0.24 (−0.30, −0.18)	<.001
2023–2024	2	Same model, map removed	−0.06 (−0.12, −0.02)	.02
		LSTM	−0.05 (−0.10, −0.01)	.03
		Gradient-boosted trees	−0.07 (−0.13, −0.02)	.03
		Seasonal ARIMA	−0.06 (−0.10, −0.03)	.005

Season	Weeks Ahead	Comparison Model	Difference (95% CI)	Adjusted P
		Seasonal blend, no learning	−0.07 (−0.12, −0.03)	.009
		Persistence	−0.25 (−0.32, −0.18)	<.001
2023–2024	4	Same model, map removed	−0.02 (−0.04, −0.00)	.09
		LSTM	−0.05 (−0.08, −0.01)	.22
		Gradient-boosted trees	−0.02 (−0.10, 0.04)	>.99
		Seasonal ARIMA	0.04 (0.01, 0.10)	.44
		Seasonal blend, no learning	0.01 (−0.02, 0.06)	>.99
		Persistence	−0.22 (−0.33, −0.12)	.002
2024–2025	1	Same model, map removed	−0.08 (−0.13, −0.05)	<.001
		LSTM	−0.06 (−0.09, −0.02)	<.001
		Gradient-boosted trees	−0.13 (−0.22, −0.06)	<.001
		Seasonal ARIMA	−0.16 (−0.27, −0.08)	<.001
		Seasonal blend, no learning	−0.20 (−0.31, −0.12)	<.001
		Persistence	−0.27 (−0.33, −0.21)	<.001
2024–2025	2	Same model, map removed	−0.10 (−0.18, −0.04)	.007
		LSTM	−0.30 (−0.58, −0.08)	.01
		Gradient-boosted trees	−0.13 (−0.25, −0.04)	.02
		Seasonal ARIMA	−0.13 (−0.27, −0.04)	.02
		Seasonal blend, no learning	−0.19 (−0.31, −0.09)	<.001
		Persistence	−0.28 (−0.35, −0.21)	<.001
2024–2025	4	Same model, map removed	−0.07 (−0.12, −0.03)	.09
		LSTM	−0.33 (−0.65, −0.07)	.18
		Gradient-boosted trees	−0.12 (−0.23, −0.04)	.13
		Seasonal ARIMA	−0.07 (−0.22, 0.03)	.28
		Seasonal blend, no learning	−0.09 (−0.17, −0.04)	.10
		Persistence	−0.25 (−0.36, −0.13)	.002
2025–2026	1	Same model, map removed	−0.08 (−0.13, −0.05)	<.001
		LSTM	−0.07 (−0.09, −0.05)	<.001
		Gradient-boosted trees	−0.12 (−0.17, −0.09)	<.001
		Seasonal ARIMA	−0.10 (−0.13, −0.07)	<.001
		Seasonal blend, no learning	−0.15 (−0.17, −0.13)	<.001
		Persistence	−0.25 (−0.30, −0.19)	<.001
2025–2026	2	Same model, map removed	−0.07 (−0.14, −0.03)	.01
		LSTM	−0.09 (−0.11, −0.05)	<.001
		Gradient-boosted trees	−0.11 (−0.17, −0.06)	.002
		Seasonal ARIMA	−0.07 (−0.12, −0.04)	.002
		Seasonal blend, no learning	−0.12 (−0.16, −0.08)	<.001
		Persistence	−0.25 (−0.30, −0.19)	<.001
2025–2026	4	Same model, map removed	−0.04 (−0.13, 0.03)	.84
		LSTM	−0.13 (−0.22, −0.03)	.12
		Gradient-boosted trees	−0.05 (−0.10, 0.01)	.37
		Seasonal ARIMA	−0.02 (−0.06, 0.03)	.84
		Seasonal blend, no learning	−0.09 (−0.13, −0.04)	.007
		Persistence	−0.32 (−0.43, −0.21)	<.001

CI, confidence interval.

Other accuracy measures

WIS is the primary measure because it scores the whole forecast, including its range. Tables S9–S11 give the point measures often reported in earlier work. In all 3 cities the graph neural network had the lowest mean absolute error and root mean squared error at every horizon among the models in Tables S9–S11. Correlation ranks the models differently at some horizons because it ignores whether forecasts are too high or too low overall.

Table S9: Point Forecast Accuracy by Model and Horizon, 14 Neighborhoods, Boston, Massachusetts, 2022–2025

	Mean Absolute Error			Root Mean Squared Error			Correlation[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	7.85	9.63	11.93	12.88	16.99	21.99	0.84	0.76	0.61
Same model, map removed	8.76	10.10	12.57	14.44	17.56	23.13	0.80	0.73	0.57
LSTM	7.96	10.19	13.02	13.18	18.01	24.09	0.87	0.76	0.54
Cola−GNN	8.19	11.04	12.27	13.77	20.56	23.11	0.85	0.70	0.61
Gradient−boosted trees	9.21	10.82	12.99	15.56	19.23	23.73	0.80	0.73	0.57
Seasonal ARIMA	9.69	11.17	13.09	16.45	19.38	23.40	0.78	0.70	0.59
Seasonal blend, no learning	9.06	10.88	13.69	15.07	19.03	25.33	0.79	0.69	0.49
FluSight baseline	9.50	11.89	16.23	15.31	20.21	29.37	0.78	0.68	0.44
Persistence	9.48	11.89	16.22	15.30	20.21	29.37	0.78	0.68	0.44

Errors are ED visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Pearson correlation between forecast and observed rates, computed for each area over the season and averaged across areas; higher is better.

Table S10: Point Forecast Accuracy by Model and Horizon, 17 Areas, Columbus, Ohio, June 2024–May 2025

	Mean Absolute Error			Root Mean Squared Error			Correlation[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	6.02	7.32	8.62	10.17	13.21	16.82	0.83	0.74	0.62
Same model, map removed	7.04	7.65	8.71	12.41	14.41	17.23	0.81	0.75	0.64
LSTM	6.94	8.60	8.78	12.63	16.36	16.85	0.81	0.71	0.57
Cola−GNN	7.99	8.55	9.03	15.40	16.29	17.24	0.77	0.70	0.64
Gradient−boosted trees	9.66	7.67	8.70	18.95	14.47	17.34	0.60	0.75	0.65
Seasonal ARIMA	7.69	8.00	8.85	14.35	15.05	17.04	0.72	0.68	0.53
Seasonal blend, no learning	7.66	8.37	9.10	13.73	15.77	17.62	0.80	0.74	0.64
FluSight baseline	7.28	8.80	11.42	10.98	14.59	20.19	0.79	0.69	0.47
Persistence	7.28	8.80	11.42	10.98	14.59	20.19	0.79	0.69	0.47

Errors are ED visits for influenza-like illness per 100,000 residents; lower is better. wk, weeks ahead.
[a] Pearson correlation between forecast and observed rates, computed for each area over the season and averaged across areas; higher is better.

Severity alerts

Thresholds. For each neighborhood (or Columbus area), intensity thresholds come from the Moving Epidemic Method under the defaults of the `mem` R package, at the 50th, 90th and 98th

Table S11: Point Forecast Accuracy by Model and Horizon, 58 ZIP Codes, Chicago, Illinois, 2022–2026

	Mean Absolute Error			Root Mean Squared Error			Correlation[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	1.68	1.79	2.01	2.41	2.51	2.71	0.58	0.54	0.44
Same model, map removed	1.74	1.88	2.04	2.47	2.61	2.80	0.52	0.45	0.38
LSTM	1.90	2.07	2.25	2.58	2.80	3.01	0.51	0.44	0.29
Gradient−boosted trees	1.87	1.96	2.10	2.53	2.66	2.81	0.50	0.44	0.38
Seasonal ARIMA	1.83	1.90	2.03	2.53	2.60	2.72	0.52	0.50	0.44
Seasonal blend, no learning	1.86	1.92	2.04	2.62	2.70	2.84	0.50	0.47	0.41
Persistence	1.87	1.99	2.24	2.98	3.13	3.39	0.47	0.42	0.33

Errors are in influenza-like illness activity levels (1–10); lower is better. Models ran at default settings. wk, weeks ahead.

[a] Pearson correlation between forecast and observed rates, computed for each area over the season and averaged across areas; higher is better.

percentile levels CDC uses. Each test season uses only earlier, non-pandemic seasons, and data stop the day the test season starts: Boston 2022–2023 uses 2017–2018 and 2018–2019; 2023–2024 adds 2022–2023; 2024–2025 adds 2023–2024. Columbus uses 2022–2023 and 2023–2024. The 2019–2020 through 2021–2022 seasons are left out because COVID-19 truncated or suppressed them. Chicago is not included: its measure is already the health department's graded 1–10 activity level.

Scores. A week is an event when the observed rate is at or above the threshold. The probability score uses each forecast's probability of being at or above the threshold, read from its calibrated 95% interval as a normal distribution; the Brier skill score compares its mean squared error with always forecasting the pooled share of event weeks. The yes/no score alerts when the best-guess forecast is at or above the threshold; the Peirce skill score is the share of event weeks alerted minus the share of non-event weeks alerted. Both pool all neighborhood-weeks of a city's unseen seasons. Intervals for differences use the moving-block bootstrap of the main text. Code: `Code/compare_severity.py` and `Paper/scripts/make_severity.py`.

Table S12: Skill at Warning of Moderate and High Influenza Weeks, by Model, City, and Horizon, Boston, Massachusetts, and Columbus, Ohio

	Brier Skill Score (probability)						Peirce Skill Score (yes/no)					
	Moderate+			High+			Moderate+			High+		
Model	1	2	4	1	2	4	1	2	4	1	2	4
Boston, Massachusetts, 2022–2025 (3 seasons): 232 moderate and 82 high neighborhood-weeks of 2,103												
Graph neural network	0.61	0.41	0.11	0.40	0.12	−0.07	0.63	0.48	0.07	0.40	0.09	0.00
Same model, map removed	0.54	0.39	0.09	0.35	0.10	−0.07	0.62	0.45	0.06	0.32	0.10	0.00
LSTM	0.59	0.41	0.04	0.40	0.09	−0.05	0.63	0.48	0.06	0.33	0.01	0.00
Cola-GNN	0.57	0.23	0.11	0.39	0.03	−0.03	0.60	0.21	0.07	0.26	0.00	0.00
FluSight baseline	0.50	0.28	−0.16	0.32	−0.12	−0.59	0.67	0.53	0.26	0.59	0.29	0.00
Persistence	0.51	0.31	−0.09	0.37	−0.03	−0.51	0.67	0.53	0.26	0.59	0.29	0.00
Columbus, Ohio, 2024–2025: 248 moderate and 107 high neighborhood-weeks of 884												
Graph neural network	0.55	0.51	0.33	0.52	0.30	−0.04	0.68	0.65	0.35	0.51	0.36	0.02
Same model, map removed	0.51	0.48	0.31	0.38	0.20	−0.06	0.61	0.54	0.24	0.34	0.13	0.00
LSTM	0.54	0.36	0.29	0.36	−0.01	−0.04	0.66	0.32	0.42	0.16	0.00	0.02
Cola-GNN	0.46	0.37	0.30	0.02	−0.01	−0.08	0.49	0.34	0.13	0.00	0.00	0.00
FluSight baseline	0.42	0.36	0.14	0.42	0.22	−0.19	0.60	0.58	0.43	0.65	0.52	0.26
Persistence	0.43	0.37	0.17	0.44	0.26	−0.09	0.60	0.58	0.43	0.65	0.52	0.26

Column numbers are weeks ahead. Moderate+, at or above the 50th-percentile intensity threshold; High+, at or above the 90th. Scores above 0 show skill; higher is better. LSTM, long short-term memory network.

Table S13: Weeks Late in First Crossing a Neighborhood's Moderate Threshold, Median Over Area-Seasons

	Boston			Columbus		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	1	2	6	3	7	9
Same model, map removed	1	2	7	7	10	11.5
LSTM	1	3	4	8	15	5
Cola-GNN	1	3	5	14	15	8
FluSight baseline	1	2	4	1	2	4
Persistence	1	2	4	1	2	4

Positive values mean the forecast first crossed the threshold after the observed rate did; persistence is late by exactly the horizon by construction. Area-seasons in which the forecast never crossed are not included. wk, weeks ahead.

Supplementary figures

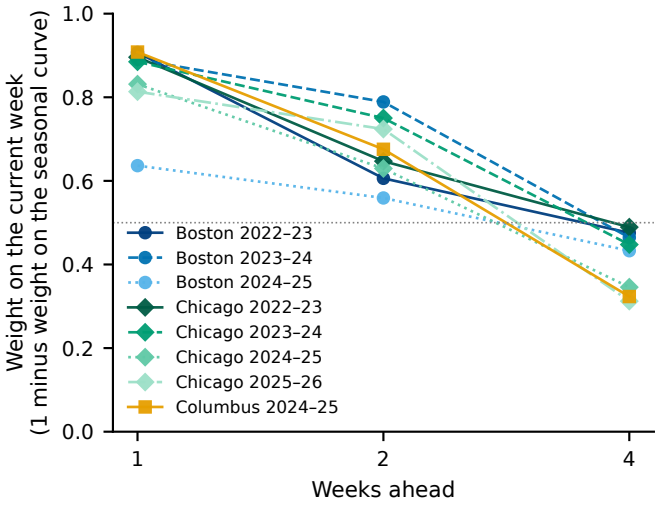


Figure S1. What the graph neural network leans on, by horizon, city, and unseen season. The model's baseline blends the current week's rate with the usual rate for that time of year; the y-axis is the learned weight on the current week (the rest goes to the seasonal curve). At 0.8 to 0.9 at 1 week in most seasons, the model mostly carries the latest value forward; by 4 weeks, it leans at least as much on the seasonal pattern. Boston, Massachusetts, 2022–2025; Columbus, Ohio, 2024–2025; Chicago, Illinois, 2022–2026 (default settings).

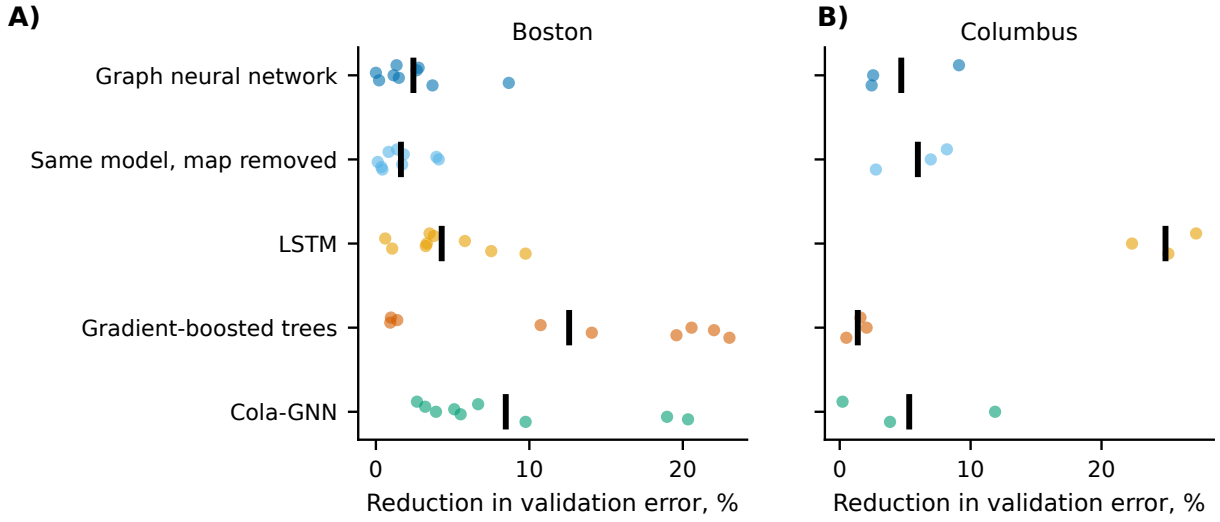


Figure S2. How much equal-budget tuning improved each model, by city: A) Boston, Massachusetts; B) Columbus, Ohio. Chicago was added after the tuning runs and is not shown. Each dot is 1 test season and horizon; the percentage is the reduction in mean absolute error on validation weeks for the chosen settings compared with the model's default settings. Black bars are means. LSTM, long short-term memory network.

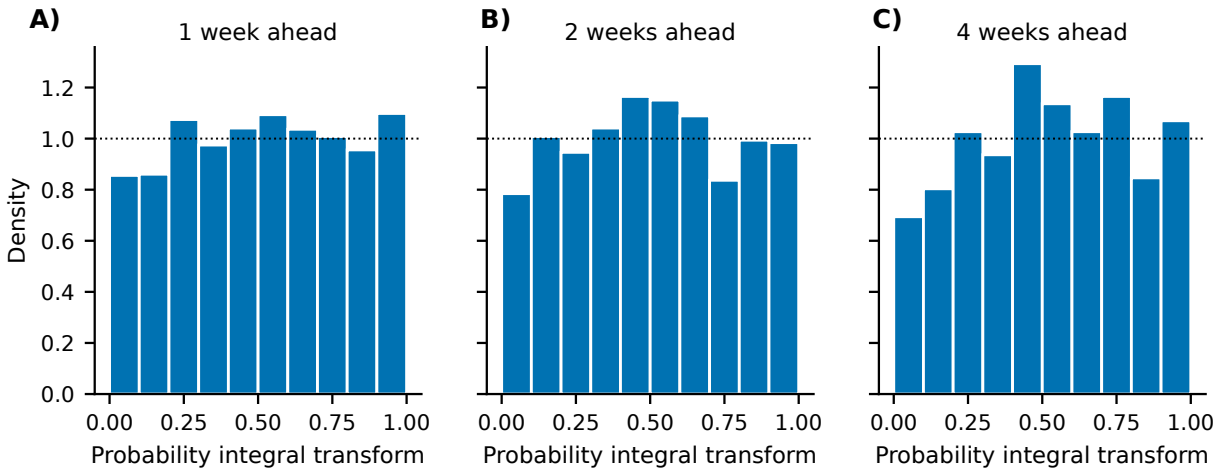


Figure S3. Calibration of the graph neural network's forecast distributions, 14 neighborhoods, Boston, Massachusetts, June 2022–May 2025: A) 1 week, B) 2 weeks, and C) 4 weeks ahead. Each histogram shows where observed values fell within the forecast distribution (the probability integral transform). A flat histogram at the dotted line means well-calibrated forecasts; a U shape would mean ranges that are too narrow, and a hump ranges that are too wide.

Data notes by city

Inputs shared by all cities. The 6 weekly weather variables are mean, maximum, and minimum temperature, relative humidity, precipitation, and maximum wind speed, from the Open-Meteo archive. The 8 demographic measures in all 3 cities are population density, poverty, share of children, share aged 65 years or older, homeownership, transit commuting, households without a vehicle, and share of nonwhite residents, from city planning tables in Boston and from American Community Survey 5-year estimates in Columbus and Chicago. Each measure is scaled within its own city.

Boston. Rates are published weekly by the Boston Public Health Commission for 14 neighborhoods (Back Bay, Beacon Hill, Downtown, North End, and West End form a single neighborhood). Suppressed weeks were treated as missing and excluded from training and scoring. Two data-loading errors found during the project (a count row averaged with the rate row, and suppressed weeks read as zero) were corrected before any result reported here was produced.

Columbus. Visit-level ED records for influenza-like illness from Columbus Public Health were assigned to 17 planning areas through the department's ZIP code crosswalk. Population denominators come from the American Community Survey 5-year estimates summed over ZIP code tabulation areas. Areas with no visits in a week are recorded as observed zeros.

Chicago. The Chicago Department of Public Health publishes a weekly influenza-like illness activity level for each ZIP code on the City of Chicago Data Portal (dataset 8vvr-jv2g), compiled from Illinois emergency department syndromic surveillance. The level compares a ZIP code's share of emergency department visits for influenza-like illness with that ZIP code's own summer average: level 1 is at or below it, each level above is roughly 1 more standard deviation, and the scale stops at 10, which 13% of ZIP-weeks reach. Because it is an index with no denominator, errors are in levels and cannot be compared in size with the other cities. The series begins in September 2019; the city is absent for May–September 2020, and those weeks are missing. Of 60 ZIP codes, 1 (60106, mostly outside the city) has no boundary and is dropped, and 1 (60827) that joined the feed in September

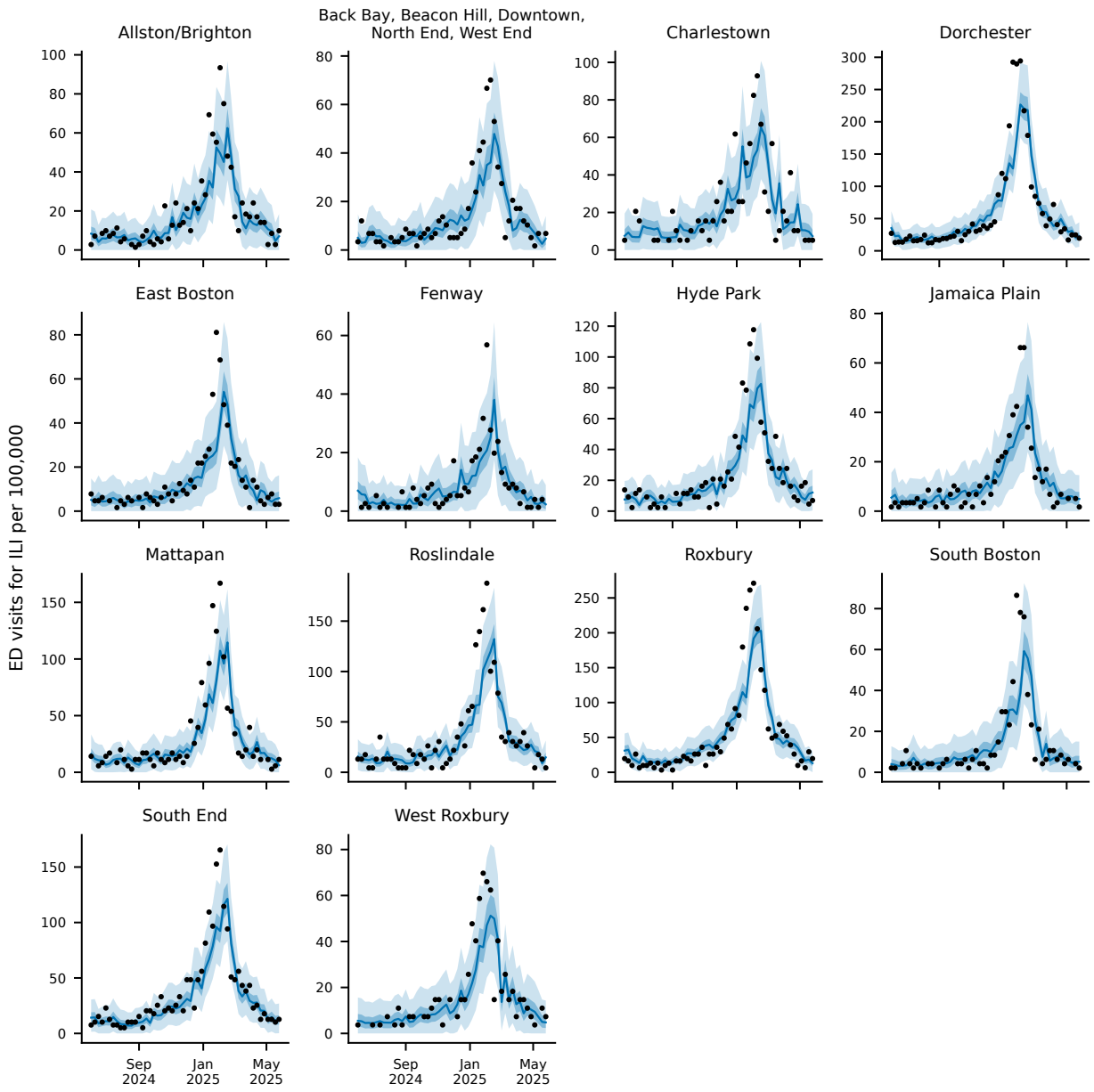


Figure S4. Forecasts made 2 weeks ahead for all 14 neighborhoods, Boston, Massachusetts, June 2024–May 2025. Lines show the graph neural network's median forecast, shading its 50% and 95% forecast intervals, and dots the observed emergency department visit rates for influenza-like illness (ILI) per 100,000 residents. Vertical scales differ by neighborhood.

2020 is in the map but not scored, leaving 58. O'Hare (60666) has no residents and takes the city median for its demographic measures. The map has 115 shared borders, derived from the city's ZIP code boundary file. Models were trained on all weeks from September 2019, and every Chicago season from 2022–2023 to 2025–2026 is unseen.

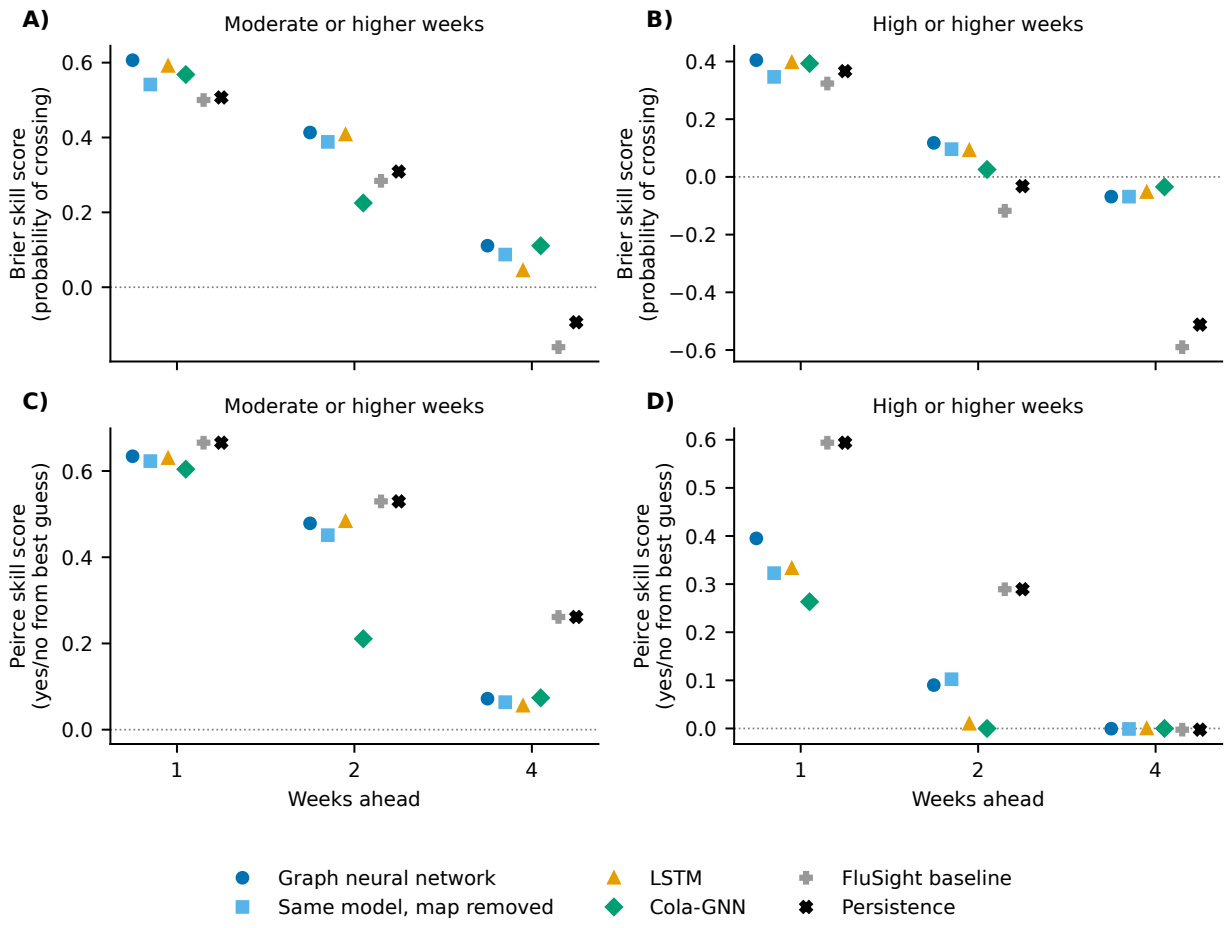


Figure S5. Skill at warning of moderate and high influenza weeks, 14 neighborhoods, Boston, Massachusetts, June 2022–May 2025. A, B) Brier skill score of the forecast's probability of being at or above the moderate (A) or high (B) threshold. C, D) Peirce skill score of a yes/no alert issued when the best-guess forecast is at or above the threshold. Values above the dotted line at 0 show skill. LSTM, long short-term memory network.

Design experience on the development season

The observations below come from the development season (2025–2026 in Boston), which was also used to choose the model's structure. They explain design choices and are not evidence of performance. Unless noted, each compares the model with one component changed against the default, averaged over 10 training runs, using the average across neighborhoods of the correlation between forecast and observed rates. The run-to-run spread of that correlation for identical settings was 0.003, 0.006, and 0.031 at 1, 2, and 4 weeks.

- **Having a map mattered; how it was wired did not.** Removing all links between neighborhoods lowered the correlation by 0.075 at 1 week and 0.093 at 2 weeks, with every neighborhood worse. Replacing the full map with shared borders only, or removing the correlation or demographic links, changed the correlation by 0.006 or less at 1 and 2 weeks.

- **Transit links did not help.** Links weighted by public transit ridership between neighbor-

hoods changed the correlation by −0.001, −0.007, and +0.015 at 1, 2, and 4 weeks. Nearly every transit link duplicated an existing link.

- **Demographic measures did not help.** Removing the 8 demographic measures changed the correlation by 0.003 or less at 1 and 2 weeks and improved it by 0.027 at 4 weeks, in all 14 neighborhoods.
- **A background node did not help.** In an earlier, untuned run of our reimplementation of the dual-topology model of Luo et al.,[3] adding a node for the surrounding area raised the correlation from 0.412 to 0.417, compared with the 0.135 gain reported in the original study.
- **Weather helped at every horizon.** Removing the 6 weather variables lowered the correlation at every horizon, and no neighborhood improved.
- **Wastewater may help at 4 weeks.** Adding Boston's influenza wastewater series raised the correlation by 0.089 at 4 weeks. The series begins in July 2024 and covers under half of the training period, so this is an upper bound; it was not part of the evaluated model.

References

[1] Deng S, Wang S, Rangwala H, et al. Cola-GNN: Cross-location Attention based Graph Neural Networks for Long-term ILI Prediction. In: Proceedings of the 29th ACM International Conference on Information & Knowledge Management. CIKM '20. New York, NY: ACM; 2020. p. 245-254. doi:10.1145/3340531.3411975.

[2] Veličković P, Cucurull G, Casanova A, et al. Graph attention networks. In: Proceedings of the 6th International Conference on Learning Representations (ICLR 2018). Vancouver, Canada; 2018. arXiv:1710.10903.

[3] Luo J, Wang X, Fan X, et al. A novel graph neural network based approach for influenza-like illness nowcasting: exploring the interplay of temporal, geographical, and functional spatial features. BMC Public Health. 2025;25(1):408. doi:10.1186/s12889-025-21618-6.